

- 1 The different structures in the human gas exchange system vary in the tissues and types of cell that are present.

Figure 1.1 is a photomicrograph showing a section of one of the structures in the human gas exchange system.

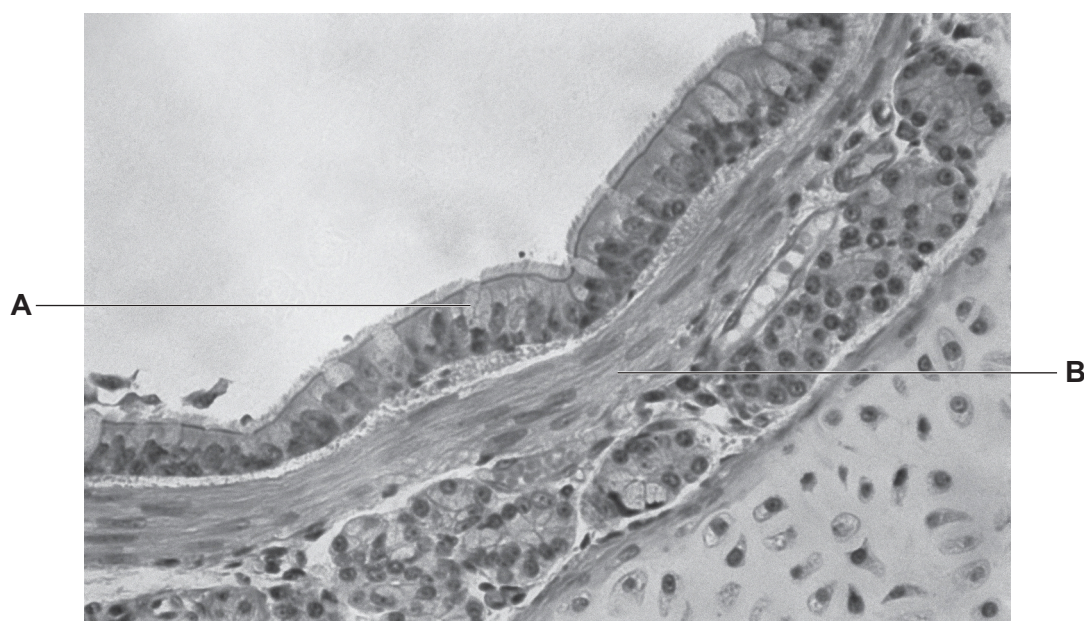


Figure 1.1

- (a) (i) Name the cell labelled **A** in Figure 1.1.

..... [1]

- (ii) Name the tissue labelled **B** in Figure 1.1.

..... [1]

- (iii) Explain why the structure in the human gas exchange system shown in Figure 1.1 is **not** a bronchiole.

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.....
.....
..... [2]

(b) Several different types of epithelial cell are present in the human gas exchange system.

These cells have different surface area to volume ratios (SA : V).

(i) A diagram of a columnar epithelial cell is shown in Figure 1.2.

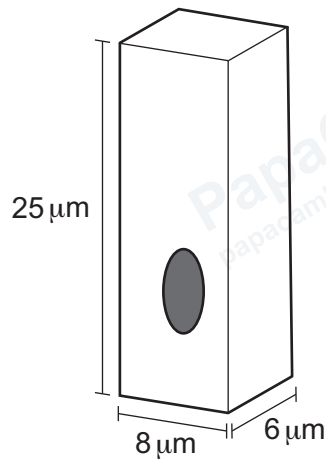


Figure 1.2

The cell in Figure 1.2 has a volume of $1200\text{ }\mu\text{m}^3$.

Calculate the SA : V of the cell shown in Figure 1.2.

Show your working **and** give the value of SA to **one** significant figure.

The value of V is 1.

SA : V : 1 [2]

(ii) A squamous epithelial cell has a typical SA : V of 10 : 1.

Explain why the high SA : V of a squamous epithelial cell is beneficial for gas exchange.

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..... [2]

[Total: 8]